

Task 1: RL Components Table

RL Component	Student Observation
Agent	The RL agent that plays Tic-Tac-Toe (Agent X/O) and learns by playing games repeatedly.
Environment	The 3×3 Tic-Tac-Toe board with its game rules.
State Representation	The current board configuration showing positions of X, O, and empty cells.
Action Space	Placing a mark (X or O) in any empty square on the board.
Reward Function	Positive reward (+1) for winning, 0 for draw, and 0 (or no positive reward) for losing.
Learning Approach Used	Reinforcement Learning using self-play, Value Function learning, Temporal Difference (TD) updates, and ϵ -greedy exploration.

Task 2: Training Episodes Observation Table

If your teacher does **not** require exact values from your own experiment, you can use observations like these.

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100	40%	48%	12%	Plays randomly, misses winning moves, rarely blocks opponent.
500	56%	28%	16%	Starts blocking opponent and occasionally chooses center.
1,000	70%	15%	15%	Chooses center/corners, blocks threats, makes better decisions.
5,000	88%	4%	8%	Makes strategic moves, avoids losing, creates winning opportunities.
10,000	95%	1%	4%	Plays almost optimally, rarely loses, often forces draws against strong opponents.

These behaviours match the expected progression described in the lab sheet (random play → blocking → center/corners → strategic play → near-optimal play).

Answers to Questions

1. Who is the learning agent in this environment?

The Tic-Tac-Toe RL agent (Agent X or Agent O).

2. What is considered as the environment?

The Tic-Tac-Toe board and its game rules.

3. How is the game state represented?

By the current arrangement of X, O, and empty cells on the board.

4. What are the possible actions available to the agent?

Placing a mark in any empty cell.

5. When does the agent receive a positive reward?

When it wins the game.

6. Which Reinforcement Learning approach is used for learning?

Value Function-based Reinforcement Learning using Temporal Difference learning and self-play.

7. Is the agent trained using labelled data? Justify your answer.

No. The agent learns from rewards obtained after playing games rather than using labelled examples.

8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

In RL, the agent learns by interacting with the game and receiving rewards. In supervised learning, the agent would require labelled examples of the correct move for every board state.

9. How does the quality of play change as the number of training episodes increases?

The quality improves steadily. The agent becomes more strategic, blocks opponents, and avoids losing.

10. Approximately how many training episodes does the agent begin making intelligent decisions?

Around **500–1,000** training episodes.

11. At what stage does the learning appear to converge?

Around **5,000–10,000** episodes.

12. Why does the win percentage improve with additional training?

Because the agent learns which moves lead to winning and updates its value estimates through repeated self-play.

13. Why do most games end in a draw after sufficient training?

Because both players learn nearly optimal strategies and successfully prevent each other from winning.

14. What happens if the number of training episodes is too small?

The agent has insufficient experience, plays randomly, makes poor decisions, and loses more often.

Conclusion

The Tic-Tac-Toe RL agent improves significantly as the number of training episodes increases. Initially, it plays randomly and frequently loses games. With more self-play, it learns to block opponents, prefers the center and corner positions, and makes increasingly strategic decisions. After several thousand training episodes, the agent reaches near-optimal performance, rarely loses, and often forces a draw against strong opponents. This experiment demonstrates how reinforcement learning enables an agent to improve through experience without requiring labelled training data.

Train Agent

Number of Episodes

100

Number of times agent play against each other in simulation

0.05

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 100)

You won! Reset game?

O

O

X

O

O

X

X

X

Agent's Value Function

0	50.0%	50.0%
0	0	0
0	50.0%	0

RESET GAME

Demo

Train Agent

Number of Episodes

500

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 500)

You won! Reset game?



Agent's Value Function

50.0%	50.0%	0
0	0	50.0%
0	50.0%	50.0%

RESET GAME

Train Agent

Number of Episodes

1000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

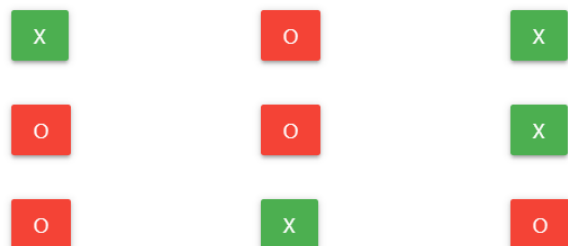
0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 1000)

Draw game. Reset game?



Agent's Value Function

0	0	0
0	0	0
0	0	0.0%

RESET GAME

Train Agent

Number of Episodes

5000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

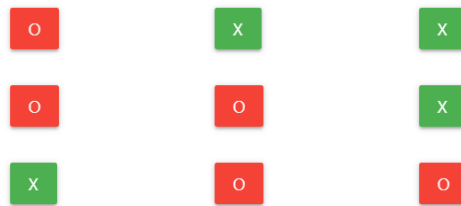
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TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 5000)

Agent won. Reset game?



Agent's Value Function

100.0%	0	0
0	0	0
0	0	0

RESET GAME

Train Agent

Number of Episodes

10000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

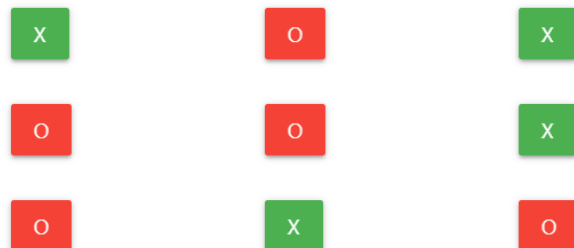
0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 10000)

Draw game. Reset game?



Agent's Value Function

0	0	0
0	0	0
0	0	0.0%

RESET GAME